

The sound of Keplerian binaries in a homogeneous Newtonian fluid background

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Abstract. We formulate the dissipation produced by sound emission from a Keplerian binary embedded in a homogeneous Newtonian fluid background. Rather than starting from the local wake force, we treat the binary as a localized periodic mass source and compute the sound-wave fluxes carried away by the medium. The energy, linear-momentum, and angular-momentum fluxes are then obtained from Noether currents of the fluid perturbations. For a classical barotropic fluid, including the self-gravity of density perturbations yields a Jeans-modified sound mode and explicit flux formulae for circular and eccentric binaries with arbitrary mass ratio. The formulation also clarifies the UV behavior of point sources: in a nondispersive fluid the harmonic sum becomes cutoff-sensitive exactly when at least one component becomes supersonic, whereas quantum pressure in a Schrödinger–Poisson quantum fluid removes this high-frequency tail and renders the fluxes UV finite. As an illustration, we translate the observed orbital decay of two binary pulsars into order-of-magnitude bounds on the homogeneous quantum-fluid density surrounding each system.

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1 Introduction

Compact binaries embedded in gaseous, superfluid, or wave-like dark-matter environments can lose orbital energy and angular momentum by exciting density perturbations in the surrounding medium. This environmental dissipation is conventionally described as *dynamical friction*: each moving body excites a wake whose gravitational field reacts back on the source. The subject originates in the classic collisionless and gaseous analyses of dynamical friction [1, 2], and has since been developed for circular motion, eccentric Keplerian motion, and binary systems in classical fluids [3–5]. Related effects have also been studied in superfluids and fuzzy or ultralight-dark-matter media, where the gravitational response may be formulated either hydrodynamically or in terms of a Schrödinger–Poisson wavefunction [6–14]. Recent work has further examined supermassive-black-hole binaries in more realistic ultralight-dark-matter halo environments [15, 16].

Most treatments are organized around the local wake force. For a prescribed periodic source in linear response, however, the same dissipative physics can be formulated as a *radiation* problem. We treat the binary as a localized periodic mass source, solve for the frequency-domain outgoing phonon field in the far zone, and obtain the energy, linear-momentum, and angular-momentum fluxes from Noether currents. This is the same strategy used in treatments of massive-field radiation [17–20]. This viewpoint complements the wake-force description: it provides the radiative field itself, places all three fluxes in a common

framework, and exposes the large-harmonic behavior directly through the phase of the outgoing wave. It also separates the kinematics of the periodic source from the dispersion and normalization of the medium mode, which makes the comparison between classical and quantum fluids particularly transparent.

We work in linear response about an infinite homogeneous Newtonian background and prescribe a Keplerian orbit, assuming that the medium-induced backreaction is weak on an orbital timescale. In the classical-fluid calculation, we retain the self-gravity of density perturbations while subtracting the gravitational field of the unperturbed infinite background. This Jeans-swindle, or local-background, prescription gives the acoustic mode a tachyonic Jeans term. We set the free Jeans modes to zero and retain only the outgoing periodically forced response. The same physical prescription is used in self-gravitating-fluid and superfluid dynamical-friction calculations [7]; here it is implemented directly at the level of the radiated field.

Starting from the fluid Lagrangian, we derive the far-zone amplitude and Noether fluxes for an arbitrary localized periodic source and then specialize to circular and eccentric binaries with arbitrary mass ratio. In the non-self-gravitating ideal-fluid limit, the fixed-center single-perturber circular result reproduces the familiar subsonic factor $\operatorname{arctanh} \mathcal{M} - \mathcal{M}$ [2, 9]. In the non-self-gravitating, fixed-center, single-perturber setup, our formulae for the energy and angular-momentum fluxes agree with the multipole sums over spherical harmonics and orbital harmonics obtained in Ref. [5] using a direct time-domain approach. The radiation formulation additionally keeps the angular emission pattern explicit, provides the linear-momentum flux, and accommodates the Jeans-modified dispersion. Its most direct qualitative consequence is a stationary-phase criterion for the point-source ultraviolet (UV) behavior: in a nondispersive classical fluid, the harmonic sum becomes logarithmically cutoff-sensitive exactly when at least one component is supersonic somewhere on the orbit.

We then repeat the construction for a Schrödinger–Poisson quantum fluid. Quantum pressure changes the high-frequency dispersion and eliminates the large-harmonic stationary point, making the point-source fluxes UV convergent. We derive the corresponding circular- and eccentric-orbit fluxes, their quadrupole limit (and the large- $\mathcal{M}_Q$ behavior in a special case). Finally, as an illustration, we derive order-of-magnitude bounds on homogeneous quantum-fluid densities from binary-pulsar orbital decay. The numerical results in this paper were produced with publicly available codes.¹

The paper is organized as follows. In Sec. 2 we formulate the radiation problem for a classical barotropic fluid, derive the Noether fluxes, and apply the result to circular and eccentric Keplerian binaries; the quadrupole and UV limits are then examined. In Sec. 3 we repeat the construction for a Schrödinger–Poisson quantum fluid. Section 4 uses binary-pulsar orbital decay to obtain illustrative bounds on homogeneous quantum-fluid densities. Appendices A and B collect frequency-domain radiation formulae for a Klein–Gordon field. Throughout this paper, we use natural units with $G = c = \hbar = 1$ and the mostly-plus metric signature.

¹The codes used to generate the results in this paper are publicly available online at <https://github.com/cao-yan-phys/binary-fluid-fluxes-calculators>.

2 Binary in a classical fluid

We consider a classical Newtonian fluid in a spatially flat Friedmann–Lemaître–Robertson–Walker (FLRW) spacetime:

$$ds^2 = -dt^2 + [a(t)]^2 |d\mathbf{x}|^2, \quad (2.1)$$

where a is the cosmological scale factor. The fluid is described by its mass density ρ , velocity $\mathbf{v}$, pressure P , and Newtonian potential Φ , which satisfy the following equations of motion:

$$\begin{aligned} \dot{\rho} + 3H\rho + \frac{\nabla \cdot (\rho \mathbf{v})}{a} &= 0, \\ \dot{\mathbf{v}} + H\mathbf{v} + \frac{\mathbf{v} \cdot \nabla \mathbf{v}}{a} &= -\frac{\nabla P}{a\rho} - \frac{\nabla \Phi}{a}, \\ \nabla^2 \Phi &= 4\pi a^2 (\rho - \bar{\rho}), \end{aligned} \quad (2.2)$$

for which a homogeneous background solution is given by $\nabla \bar{\rho} = \nabla \bar{P} = \nabla \bar{\Phi} = \bar{\mathbf{v}} = \mathbf{0}$. We are interested in the linear perturbation of this fluid background by a set of nonrelativistic point masses, corresponding to the density perturbation $\rho_* = a^{-3} \sum_I m_I \delta^3(\mathbf{x} - \mathbf{X}_I(t))$, where $\mathbf{X}_I(t)$ denotes the comoving coordinates of the I -th body with mass m_I . We decompose the fields as

$$\rho = \bar{\rho}(1 + \alpha) + \rho_*, \quad P = \bar{P} + p, \quad \Phi = \bar{\Phi} + \Phi, \quad \mathbf{v} = \bar{\mathbf{v}} + \mathbf{u}. \quad (2.3)$$

The linearized fluid equations then read

$$\begin{aligned} \nabla^2 \Phi &= 4\pi a^2 (\bar{\rho} \alpha + \rho_*), \\ \dot{\mathbf{u}} + H\mathbf{u} &= -\frac{\nabla p}{a\bar{\rho}} - \frac{\nabla \Phi}{a}, \\ \dot{\alpha} + \frac{\nabla \cdot \mathbf{u}}{a} &= 0. \end{aligned} \quad (2.4)$$

We assume that the fluid is barotropic, with squared sound speed $c_s^2 \equiv p/(\bar{\rho} \alpha)$. We refer to this model below as the “classical fluid”. It follows that

$$\ddot{\alpha} + 2H\dot{\alpha} - c_s^2 \frac{\nabla^2 \alpha}{a^2} = 4\pi (\bar{\rho} \alpha + \rho_*). \quad (2.5)$$

A time-dependent source $\rho_*(t, \mathbf{x})$ thus generates sound waves described by the “phonon field” α . Note that the first term on the right-hand side of Eq. (2.5) arises from the self-gravity of the fluid. Henceforth we consider the sub-horizon limit with $a = 1$, so the fluid equations reduce to

$$\begin{aligned} \nabla^2 \Phi &= 4\pi (\bar{\rho} \alpha + \rho_*), \\ \dot{\mathbf{u}} &= -\frac{\nabla p}{\bar{\rho}} - \nabla \Phi = -c_s^2 \nabla \alpha - \nabla \Phi, \\ \dot{\alpha} + \nabla \cdot \mathbf{u} &= 0, \end{aligned} \quad (2.6)$$

and

$$\ddot{\alpha} - c_s^2 \nabla^2 \alpha = 4\pi (\bar{\rho} \alpha + \rho_*). \quad (2.7)$$

We now decompose the velocity perturbation as $\mathbf{u} = \nabla s + \mathbf{U}$, $\nabla \cdot \mathbf{U} = 0$. The equations become:

$$\begin{aligned} \nabla^2 \Phi &= 4\pi (\bar{\rho} \alpha + \rho_*), \\ \nabla \dot{s} + \dot{\mathbf{U}} &= -c_s^2 \nabla \alpha - \nabla \Phi, \\ \dot{\alpha} + \nabla^2 s &= 0. \end{aligned} \quad (2.8)$$

We can write $\dot{\mathbf{U}} = \nabla\chi$. Since $\nabla \cdot \mathbf{U} = 0$, it follows that $\nabla \cdot \dot{\mathbf{U}} = 0$, and therefore $\nabla^2\chi = 0$. Assuming that χ vanishes, or approaches a constant, at spatial infinity, the uniqueness theorem for Laplace's equation implies that $\chi = 0$ up to an irrelevant constant. Therefore, $\dot{\mathbf{U}} = 0$, and $\dot{s} = -c_s^2\alpha - \Phi + C(t)$. Since $C(t)$ can be absorbed into a redefinition of s , we choose the gauge:

$$\dot{s} = -c_s^2\alpha - \Phi. \quad (2.9)$$

2.1 Sound waves

We proceed to analyze some consequences of Eqs. (2.6). Defining $\nabla^2\Phi_* = 4\pi\rho_*$ and $\nabla^2\Phi_f = 4\pi\bar{\rho}\alpha$, it follows from Eqs. (2.6) that

$$\partial_t \left(\frac{1}{2}\bar{\rho}u^2 + \frac{1}{2}c_s^2\bar{\rho}\alpha^2 + \frac{1}{8\pi}|\nabla\Phi_f|^2 + \bar{\rho}\alpha\Phi_f \right) + \nabla \cdot \left(p\mathbf{u} - \frac{1}{4\pi}\dot{\Phi}_f \nabla\Phi_f + \bar{\rho}\Phi_f\mathbf{u} \right) = -\bar{\rho}\mathbf{u} \cdot \nabla\Phi_*, \quad (2.10)$$

we can thus identify

$$\mathcal{E} = \frac{1}{2}\bar{\rho}u^2 + \frac{1}{2}c_s^2\bar{\rho}\alpha^2 + \frac{1}{8\pi}|\nabla\Phi_f|^2 + \bar{\rho}\alpha\Phi_f \quad (2.11)$$

as the energy density, and

$$\mathbf{S} = p\mathbf{u} - \frac{1}{4\pi}\dot{\Phi}_f \nabla\Phi_f + \bar{\rho}\Phi_f\mathbf{u} \quad (2.12)$$

as the energy flux density. For $\rho_* = \sum_I m_I \delta^3(\mathbf{x} - \mathbf{X}_I(t))$, the right-hand side of Eq. (2.10) is the power delivered by the external gravitational field of the point masses to the fluid perturbation, since

$$\int d^3x (-\bar{\rho}\mathbf{u} \cdot \nabla\Phi_*) = \sum_I m_I \mathbf{v}_I \cdot \left[\int d^3x \bar{\rho}\alpha \frac{\mathbf{x} - \mathbf{X}_I(t)}{|\mathbf{x} - \mathbf{X}_I(t)|^3} \right]. \quad (2.13)$$

We assume that the source is prescribed and periodic, with period T . This approximation is appropriate when the fluid perturbation is weak and on timescales short compared with the backreaction time. We also set the free (homogeneous) modes of the α field to zero. The outgoing sound-wave flux is therefore supplied entirely by the source, yielding the corresponding energy-balance relation.

To simplify the analysis, we now turn to a Lagrangian formulation,² which makes the derivation of the radiation fluxes more convenient. The Lagrangian for $\{s, \alpha, \Phi\}$ can be taken as

$$\mathcal{L} = -\frac{\bar{\rho}}{2}|\nabla s|^2 - \frac{\bar{\rho}}{2}c_s^2\alpha^2 - \bar{\rho}\Phi\alpha + \bar{\rho}s\dot{\alpha} - \frac{1}{8\pi}|\nabla\Phi|^2 - \rho_*\Phi. \quad (2.14)$$

From Noether's theorem (setting $\rho_* = 0$), we obtain the energy flux density:

$$\mathbf{S} = \frac{\partial\mathcal{L}}{\partial\nabla s}\dot{s} + \frac{\partial\mathcal{L}}{\partial\nabla\Phi}\dot{\Phi} = -\bar{\rho}\dot{s}\nabla s - \frac{1}{4\pi}\dot{\Phi}\nabla\Phi, \quad (2.15)$$

the energy density:

$$\mathcal{E} = \frac{\partial\mathcal{L}}{\partial\dot{\alpha}}\dot{\alpha} - \mathcal{L} = \frac{\bar{\rho}}{2}|\nabla s|^2 + \frac{1}{8\pi}|\nabla\Phi|^2 + \frac{\bar{\rho}}{2}c_s^2\alpha^2 + \bar{\rho}\Phi\alpha, \quad (2.16)$$

²In view of Eq. (2.7), one might try to construct a phonon Lagrangian $\mathcal{L}_\alpha = A \left[\frac{1}{2}(\dot{\alpha})^2 - \frac{1}{2}c_s^2|\nabla\alpha|^2 + \frac{1}{2}(4\pi\bar{\rho})\alpha^2 + 4\pi\rho_*\alpha \right]$, and derive the corresponding Noether currents. This indeed gives the correct results if the constant factor A is chosen appropriately; see Sec. 3.3 for a discussion.

These expressions reproduce Eqs. (2.11)-(2.12). The linear momentum flux density is

$$\Pi_q^i = - \left[\frac{\partial \mathcal{L}}{\partial(\partial_i s)} \partial_q s + \frac{\partial \mathcal{L}}{\partial(\partial_i \Phi)} \partial_q \Phi - \delta_{iq} \mathcal{L} \right] = \bar{\rho} \partial_i s \partial_q s + \frac{1}{4\pi} \partial_i \Phi \partial_q \Phi + \delta_{iq} \mathcal{L}, \quad (2.17)$$

the linear momentum density:

$$p_q = - \frac{\partial \mathcal{L}}{\partial \dot{\alpha}} \partial_q \alpha = - \bar{\rho} s \partial_q \alpha, \quad (2.18)$$

the angular momentum flux density:

$$\begin{aligned} j_q^i &= \frac{1}{2} \epsilon_{qkl} \left[\frac{\partial \mathcal{L}}{\partial(\partial_i s)} A_{kl}^j \partial_j s + \frac{\partial \mathcal{L}}{\partial(\partial_i \Phi)} A_{kl}^j \partial_j \Phi - A_{kl}^i \mathcal{L} \right] \\ &= \epsilon_{qkl} x^l \left(- \bar{\rho} \partial_i s \partial_k s - \frac{1}{4\pi} \partial_i \Phi \partial_k \Phi - \delta_{ik} \mathcal{L} \right), \end{aligned} \quad (2.19)$$

and the angular momentum density:

$$j_q = \frac{1}{2} \epsilon_{qkl} \left(\frac{\partial \mathcal{L}}{\partial \dot{\alpha}} A_{kl}^j \partial_j \alpha \right) = \bar{\rho} \epsilon_{qkl} x^l s \partial_k \alpha. \quad (2.20)$$

2.2 Radiation from a localized periodic source

We proceed to evaluate the radiation fluxes for a localized periodic sound source. To this end, we need a general expression for the sound waves. For convenience, we introduce an auxiliary time coordinate $\mathfrak{t} \equiv c_s t$, so that Eq. (2.7) becomes

$$\partial_{\mathfrak{t}}^2 \alpha - \nabla^2 \alpha - m^2 \alpha = n(\mathfrak{t}, \mathbf{x}) \equiv \frac{4\pi}{c_s^2} \rho_*(t = \mathfrak{t}/c_s, \mathbf{x}), \quad (2.21)$$

where the effective tachyonic mass is defined by

$$m \equiv \frac{\sqrt{4\pi\bar{\rho}}}{c_s}. \quad (2.22)$$

The dispersion relation of a free plane wave $\alpha \propto e^{i(\mathbf{k} \cdot \mathbf{x} - \omega \mathfrak{t})}$ is $\omega^2 = |\mathbf{k}|^2 - m^2$, which exhibits a Jeans instability for $|\mathbf{k}| < k_J \equiv \sqrt{4\pi\bar{\rho}}/c_s$. This instability, however, does not affect the forced radiative solution: the source fixes the wave frequency, for which $|\mathbf{k}| = \sqrt{\omega^2 + m^2}$. The free Jeans modes are set to zero by the imposed boundary conditions, leaving only the forced linear response of the medium.

In the frequency domain, Eq. (2.21) reads $(\nabla^2 + m^2 + \omega^2) \tilde{\alpha} = -\tilde{n}$, with

$$\tilde{\alpha}(\omega, \mathbf{x}) = \int_{-\infty}^{\infty} d\mathfrak{t} \alpha(\mathfrak{t}, \mathbf{x}) e^{i\omega \mathfrak{t}}, \quad \tilde{n}(\omega, \mathbf{x}) = \int_{-\infty}^{\infty} d\mathfrak{t} n(\mathfrak{t}, \mathbf{x}) e^{i\omega \mathfrak{t}}. \quad (2.23)$$

For $\omega > 0$, the outgoing solution to $(\nabla^2 + m^2 + \omega^2) f(\mathbf{x}, \mathbf{x}') = \delta^3(\mathbf{x} - \mathbf{x}')$ is

$$f(\mathbf{x}, \mathbf{x}') = \int \frac{d^3 k}{(2\pi)^3} \frac{e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')}}{-|\mathbf{k}|^2 + m^2 + (\omega + i0^+)^2} = - \frac{e^{i\sqrt{\omega^2 + m^2} |\mathbf{x} - \mathbf{x}'|}}{4\pi |\mathbf{x} - \mathbf{x}'|} \quad (2.24)$$

The negative-frequency branch in the two-sided representation is obtained by complex conjugation. Therefore, for a source with period $T \equiv 2\pi/\tilde{\Omega}$, the radiation field is [20]

$$\alpha(\mathbf{t}, \mathbf{x} = R \mathbf{n}) = \frac{1}{4\pi R} \sum_n n_{(n)} e^{i[\mathbf{k}^{(n)} \cdot \mathbf{x} - \Omega_n \mathbf{t}]}, \quad (2.25)$$

$$n_{(n)} = \int d^3x e^{-i\mathbf{k}^{(n)} \cdot \mathbf{x}} \left[\frac{1}{T} \int_0^T d\mathbf{t} n(\mathbf{t}, \mathbf{x}) e^{i\Omega_n \mathbf{t}} \right], \quad (2.26)$$

where $T \equiv 2\pi/\Omega = c_s T$, $\Omega_n \equiv n\Omega = n\tilde{\Omega}/c_s$, $\mathbf{k}^{(n)} = k_n \mathbf{n} = \Omega_n \sqrt{1 + (n_0/n)^2} \mathbf{n}$ (note that $k_{-n} = -k_n$), and $n_0 \equiv m/\Omega = \sqrt{4\pi\bar{\rho}}/\tilde{\Omega}$. See also Appendix A for a time-domain derivation. The radiation field can also be expanded in spherical harmonics by evaluating the integral in Eq. (2.26) with $\mathbf{n} = (\theta, \phi)$ and $\mathbf{x} = r' \mathbf{n}'(\theta', \phi')$ using the Rayleigh expansion:

$$e^{-i\mathbf{k}^{(n)} \cdot \mathbf{x}} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l (-i)^l j_l^*(k_n r') Y_{lm}^*(\theta', \phi') Y_{lm}(\theta, \phi).$$

For the radiation field, we also have

$$\begin{pmatrix} \Phi \\ s \end{pmatrix} = \frac{1}{4\pi R} \sum_n \begin{pmatrix} \kappa_n^\Phi \\ \kappa_n^s \end{pmatrix} n_{(n)} e^{i[\mathbf{k}^{(n)} \cdot \mathbf{x} - \Omega_n \mathbf{t}]}, \quad (2.27)$$

with

$$\kappa_n^\Phi = \frac{-4\pi\bar{\rho}}{k_n^2}, \quad \kappa_n^s = \frac{-ic_s\Omega_n}{k_n^2}. \quad (2.28)$$

Using Eqs. (2.25) and (2.27), the time-averaged energy flux is

$$P = R^2 \int d\Omega_{\mathbf{n}} \mathbf{S} \cdot \mathbf{n} = \int d\Omega_{\mathbf{n}} \sum_{n \geq 1} \frac{\bar{\rho} c_s^3 \Omega_n |n_{(n)}|^2}{8\pi^2 k_n}. \quad (2.29)$$

The time-averaged energy density is given by

$$\mathcal{E} = \frac{\bar{\rho}}{2} |\nabla s|^2 + \frac{1}{8\pi} |\nabla \Phi|^2 + \frac{\bar{\rho}}{2} c_s^2 \alpha^2 + \bar{\rho} \Phi \alpha = \frac{1}{R^2} \sum_{n \geq 1} \frac{\bar{\rho} c_s^2 \Omega_n^2}{8\pi^2 k_n^2} |n_{(n)}|^2. \quad (2.30)$$

As expected, this quantity satisfies $\mathbf{S} \cdot \mathbf{n} = \mathcal{E} v_g$ mode by mode, where $v_g = c_s k_n / \Omega_n$ is the group velocity of the sound wave. The time-averaged linear momentum density is

$$p_q = -\bar{\rho} s \partial_q \alpha = \sum_{n \geq 1} \frac{\bar{\rho} c_s \Omega_n}{8\pi^2 k_n R^2} |n_{(n)}|^2 n^q, \quad (2.31)$$

so the time-averaged linear momentum flux is

$$\mathcal{F}_q = \sum_{n \geq 1} \int d\Omega_{\mathbf{n}} \frac{\bar{\rho} c_s^2}{8\pi^2} |n_{(n)}|^2 n^q, \quad (2.32)$$

which is consistent with the direct evaluation: $\mathcal{F}_q = R^2 \int d\Omega_{\mathbf{n}} \Pi_q^i n^i$. The time-averaged angular momentum density is

$$j_q = \bar{\rho} \epsilon_{qkl} x^l s \partial_k \alpha = \sum_n (-i) \frac{\bar{\rho} c_s \Omega_n}{16\pi^2 k_n^2 R^2} \epsilon_{qkl} x^l n_{(n)} \partial_k n_{(n)}^*. \quad (2.33)$$

Using $n_{(-n)}(\mathbf{n}) = n_{(n)}^*(\mathbf{n})$, the time-averaged angular momentum flux is

$$\begin{aligned}\tau_q &= \int d\Omega_{\mathbf{n}} \sum_{n \neq 0} (-i) \frac{\bar{\rho} c_s^2}{16\pi^2 k_n} \epsilon_{qkl} x^l n_{(n)} \partial_k n_{(n)}^* \\ &= \text{Re} \int d\Omega_{\mathbf{n}} \sum_{n \geq 1} (-i) \frac{\bar{\rho} c_s^2}{8\pi^2 k_n} \epsilon_{qkl} x^l n_{(n)} \partial_k n_{(n)}^*.\end{aligned}\quad (2.34)$$

This result is consistent with the direct evaluation, $\tau_q = R^2 \int d\Omega_{\mathbf{n}} j_q^i n^i$.

Because the effective mass is tachyonic, the radiation contains a zero-frequency mode with $k_n = \pm m$. For point sources $\rho_* = a^{-3} \sum_I m_I \delta^3(\mathbf{x} - \mathbf{X}_I(t))$,

$$n_{(\pm 0)}(\mathbf{n}) \propto \sum_I m_I \left(\frac{1}{T} \int_0^T dt e^{\mp i k_n \mathbf{n} \cdot \mathbf{X}_I} \right). \quad (2.35)$$

It follows that $n_{(\pm 0)}(-\mathbf{n}) = n_{(\pm 0)}^*(\mathbf{n}) = n_{(\mp 0)}(\mathbf{n})$, so $|n_{(\pm 0)}(-\mathbf{n})|^2 = |n_{(\pm 0)}(\mathbf{n})|^2$, and

$$\partial_k n_{(\pm 0)}(\mathbf{n}) \propto \sum_I m_I \left[\frac{\mp i k_n}{T} \int_0^T dt e^{\mp i k_n \mathbf{n} \cdot \mathbf{X}_I} \frac{X_I^k - n^k (\mathbf{n} \cdot \mathbf{X}_I)}{R} \right] = -\partial_k n_{(\mp 0)}(-\mathbf{n}). \quad (2.36)$$

The zero-frequency mode therefore does not contribute to the radiation fluxes. The nonzero tachyonic mass is essential here because it removes the infrared (IR) divergence that would otherwise be associated with the zero-frequency mode.

Having obtained the general expressions for the radiation fluxes, we now specialize to a Keplerian binary and derive more explicit results. Since the unperturbed homogeneous fluid background does not affect the binary motion, assuming a fixed Keplerian orbit amounts to requiring that the fluid perturbation induced by the binary be sufficiently weak.

2.3 Circular orbit

For a Keplerian binary with $\rho_* = \sum_{I=1,2} m_I \delta^3(\mathbf{x} - \mathbf{X}_I(t))$, we choose a Cartesian coordinate system in which the z -axis aligns with the binary's orbital angular momentum and the coordinate origin is at the binary's center of mass. We denote the total mass of the binary by $M \equiv m_1 + m_2$, and the symmetric mass ratio by $\nu \equiv m_1 m_2 / M^2$.

We begin with the case of a circular orbit,³ and set

$$\begin{aligned}\mathbf{X}_1 &= R_1 (\cos \Omega t, \sin \Omega t, 0), \\ \mathbf{X}_2 &= -R_2 (\cos \Omega t, \sin \Omega t, 0), \\ \mathbf{k}^{(n)} &= k_n \mathbf{n}, \\ \mathbf{n} &= (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).\end{aligned}\quad (2.37)$$

The two-body separation is $a = R_1 + R_2$, and the velocity of the I -th body is denoted by $\mathbf{v}_I \equiv \dot{\mathbf{X}}_I$. The Bessel function of the first kind has the integral representation

$$J_n(z) = \int_0^{2\pi} \frac{d\xi}{2\pi} e^{i(n\xi - z \sin \xi)} \quad (2.38)$$

³In this case, it is not necessary to assume a Keplerian orbit, so the result is more broadly applicable.

Equation (2.26) then gives

$$\begin{aligned} \frac{n(n)}{4\pi/c_s^2} &= \sum_{I=1,2} \frac{m_I}{T} \int_0^T dt e^{i\Omega_n t} e^{-i\mathbf{k}^{(n)} \cdot \mathbf{X}_I} \\ &= e^{in(\phi-\pi/2)} [m_1 J_n(k_n R_1 \sin \theta) + (-1)^n m_2 J_n(k_n R_2 \sin \theta)]_{\underline{+}}. \end{aligned} \quad (2.39)$$

From Eq. (2.29), the energy flux is given by

$$P = \frac{4\pi\bar{\rho}M^2}{c_s} \sum_{n \geq 1} \frac{\Omega_n}{k_n} I_n \equiv \frac{2\bar{\rho}M^2}{c_s} \mathcal{X}, \quad (2.40)$$

where

$$I_n \equiv \int \sin \theta d\theta \left[\frac{m_1}{M} J_n(k_n R_1 \sin \theta) + (-1)^n \frac{m_2}{M} J_n(k_n R_2 \sin \theta) \right]^2. \quad (2.41)$$

Some concrete examples are shown in Fig. 1. In the special case of equal masses, $m_1 = m_2 = M/2$, so that $R_1 = R_2 = a/2$, I_n is nonzero only if n is even, and can be obtained analytically as

$$\begin{aligned} I_n &= \int \sin \theta d\theta \left[J_n \left(\frac{1}{2} k_n a \sin \theta \right) \right]^2 \\ &= \left(\frac{k_n a}{2} \right)^{2n} \Gamma \left(n + \frac{1}{2} \right) {}_1\tilde{F}_2 \left(n + \frac{1}{2}; n + \frac{3}{2}, 2n + 1; - \left(\frac{k_n a}{2} \right)^2 \right), \end{aligned} \quad (2.42)$$

where ${}_1\tilde{F}_2(\mathbf{a}; \mathbf{b}; z)$ is the regularized generalized hypergeometric function. We define the Mach number $\mathcal{M} \equiv a\Omega$. The velocity of each body is $v = a\Omega/2$; hence $\mathcal{M} = 2v/c_s$. For $m = 0$,⁴ $k_n a = n\mathcal{M}$. In the subsonic regime ($\mathcal{M} < 2$), the infinite series turns out to be

$$\sum_{n=1}^{\infty} I_n(m=0) = \sum_{n=\text{even}} I_n = \frac{\operatorname{arctanh} \frac{\mathcal{M}}{2} + \ln(\tan \frac{\mathcal{E}}{2})}{\mathcal{M}} - 1, \quad (2.43)$$

where $\mathcal{E} > 0$ is the solution of $\mathcal{E} - \frac{\mathcal{M}}{2} \sin \mathcal{E} = \frac{\pi}{2}$. Therefore, we obtain

$$F_{\text{tangent}}(m=0) = \frac{P}{2v} = \frac{4\pi\bar{\rho}M^2}{v^2} \sum_{n \geq 1} \frac{\mathcal{M}}{4} I_n = \frac{4\pi\bar{\rho}M^2}{v^2} \left[\frac{\operatorname{arctanh} \frac{\mathcal{M}}{2} + \ln(\tan \frac{\mathcal{E}}{2})}{4} - \frac{\mathcal{M}}{4} \right]. \quad (2.44)$$

For $\mathcal{M} > 2$, the infinite series diverges. To see this, consider the large- n limit, in which we may set $n_0 = 0$; then $\lim_{n \rightarrow \infty} I_n \rightarrow \frac{2}{n\mathcal{M}}$. Therefore,

$$\lim_{n_{\max} \rightarrow \infty} \sum_{n=\text{even}}^{n_{\max}} I_n \sim \frac{1}{\mathcal{M}} \sum_{n=1}^{n_{\max}} \frac{1}{n} \sim \frac{\ln n_{\max}}{\mathcal{M}}. \quad (2.45)$$

The energy flux thus exhibits a logarithmic UV divergence. In reality, it should be regularized by a physical cutoff $n_{\max} \sim c_s/(\bar{\Omega} R_{\text{eff}})$, with R_{eff} representing the finite body size or another microscopic smoothing scale. For unequal masses, the threshold for UV divergence can be shown to be

$$\frac{1 + \sqrt{1 - 4\nu}}{2} \mathcal{M} \geq 1. \quad (2.46)$$

⁴For $m = 0$, we exclude the zero-frequency ($n = 0$) mode. This amounts to considering a negligible but non-vanishing n_0 , so that the result is still free of IR divergence.

Equivalently, this reads

$$\max(|\mathbf{v}_1|, |\mathbf{v}_2|) = \frac{1 + \sqrt{1 - 4\nu}}{2} |\mathbf{v}_1 - \mathbf{v}_2| \geq c_s. \quad (2.47)$$

The same supersonic threshold applies to eccentric orbits, as shown in the next subsection. An analytic expression for the energy flux is difficult to obtain when the self-gravity of the fluid is included ($n_0 > 0$). In the examples shown in Fig. 1, self-gravity increases the energy flux.

In the fixed-center single-perturber setup, body 1 is held fixed and body 2 (with mass νM) moves on a circular orbit of radius a with speed $v = a\tilde{\Omega}$.⁵ The corresponding Mach number is $\mathcal{M} = v/c_s = a\Omega$. For $m = 0$,

$$I_n(m = 0) = \nu^2 \int_0^\pi d\theta \sin \theta J_n^2(n\mathcal{M} \sin \theta), \quad (2.48)$$

and the normalized sum gives

$$\sum_{n=1}^{\infty} I_n(m = 0) = \nu^2 \left(\frac{\operatorname{arctanh} \mathcal{M}}{\mathcal{M}} - 1 \right), \quad (2.49)$$

and thus

$$F_{\text{tangent}} = \frac{4\pi\bar{\rho}(\nu M)^2}{v^2} (\operatorname{arctanh} \mathcal{M} - \mathcal{M}). \quad (2.50)$$

See Fig. 1 for a concrete example. The dimensionless factor $\operatorname{arctanh} \mathcal{M} - \mathcal{M} = \frac{1}{2} \ln[(1 + \mathcal{M})/(1 - \mathcal{M})] - \mathcal{M}$ is identical to the finite subsonic gaseous-drag factor of the straight-line problem [2]. The same result can be obtained by the direct time-domain calculation [5, 9], which provides a nontrivial cross-check of the present flux-based approach. The absence of a large-distance logarithm in this subsonic limit should be contrasted with the logarithmic sensitivity familiar from collisionless dynamical friction and from supersonic straight-line gaseous wakes [1, 2, 6]. In addition to the tangential gravitational force exerted by the fluid perturbation, the time-domain calculation can reveal the force perpendicular to the velocity, which cannot be reconstructed from the global radiation fluxes alone, although it does not enter the energy balance and therefore does not affect the leading dissipative adiabatic evolution inferred from the radiated power. Extending a direct wake calculation beyond the fixed-center single-perturber setup is comparatively cumbersome. The radiation formulation, by contrast, extends directly to eccentric orbits with arbitrary mass ratio.

2.4 Eccentric orbit

For a Keplerian eccentric orbit, we can set $\mathbf{X}_1 = \frac{m_2}{M} \mathbf{X}$ and $\mathbf{X}_2 = -\frac{m_1}{M} \mathbf{X}$, with the parameterization:

$$\begin{aligned} \mathbf{X} &\equiv \mathbf{X}_1 - \mathbf{X}_2 = a \left(\cos \xi - e, \sqrt{1 - e^2} \sin \xi, 0 \right), \\ \Omega \mathbf{t} &= \xi - e \sin \xi. \end{aligned} \quad (2.51)$$

⁵This setup is not exactly the same as the true two-body extreme-mass-ratio limit $m_2/m_1 \ll 1$, where the heavier body still moves about the center of mass. See Fig. 3 for a direct comparison of the radiation fluxes from an eccentric orbit.

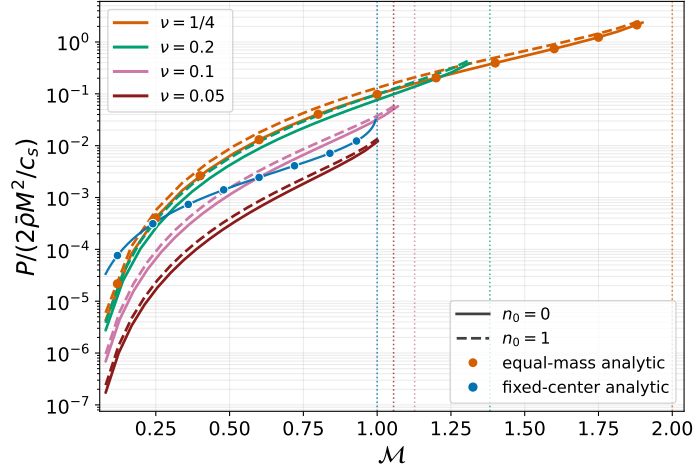


Figure 1. The dimensionless circular-orbit function $\mathcal{X}(\mathcal{M})$ for different symmetric mass ratios. Solid and dashed curves correspond to $n_0 = 0$ and $n_0 = 1$. The points show the equal-mass and fixed-center analytic results, given by Eqs. (2.43) and (2.49), respectively. The vertical dotted lines mark the supersonic threshold (2.47).

We choose $m_1 \geq m_2$, so that $m_1/M = (1 + \sqrt{1 - 4\nu})/2$ and $m_2/M = (1 - \sqrt{1 - 4\nu})/2$. The source amplitude in Eq. (2.26) evaluates to

$$\begin{aligned} n_{(n)} &= \frac{4\pi}{c_s^2} \int_0^{2\pi} \frac{d\xi}{2\pi} (1 - e \cos \xi) e^{in(\xi - e \sin \xi)} \left(m_1 e^{-i \frac{m_2}{M} k_n \cdot \mathbf{n} \cdot \mathbf{X}} + m_2 e^{i \frac{m_1}{M} k_n \cdot \mathbf{n} \cdot \mathbf{X}} \right) \\ &= \frac{4\pi M}{c_s^2} K_n, \end{aligned} \quad (2.52)$$

with

$$K_n \equiv \int_0^{2\pi} \frac{d\xi}{2\pi} (1 - e \cos \xi) e^{in(\xi - e \sin \xi)} \left[\frac{m_1}{M} e^{-i \frac{m_2}{M} (ak_n) \cdot \mathbf{n} \cdot (\mathbf{X}/a)} + \frac{m_2}{M} e^{i \frac{m_1}{M} (ak_n) \cdot \mathbf{n} \cdot (\mathbf{X}/a)} \right], \quad (2.53)$$

where $n_0 \equiv m/\Omega$. From Eq. (2.29), the energy flux is given by

$$P = \frac{2\bar{\rho}M^2}{c_s} \sum_{n \geq 1} \frac{\int \sin \theta d\theta d\phi |K_n|^2}{\sqrt{1 + (n_0/n)^2}}. \quad (2.54)$$

From Eq. (2.34), the angular momentum flux is given by

$$\begin{aligned} \tau_q \tilde{\Omega} &= \frac{2\bar{\rho}M^2}{c_s} \sum_{|n| \geq 1} \frac{(-i)/(2n)}{\sqrt{1 + (n_0/n)^2}} \int \sin \theta d\theta d\phi \begin{pmatrix} \sin \phi K_n \partial_\theta K_n^* + \cot \theta \cos \phi K_n \partial_\phi K_n^* \\ -\cos \phi K_n \partial_\theta K_n^* + \cot \theta \sin \phi K_n \partial_\phi K_n^* \\ -K_n \partial_\phi K_n^* \end{pmatrix} \\ &= \frac{2\bar{\rho}M^2}{c_s} \text{Re} \sum_{n \geq 1} \frac{(-i)/n}{\sqrt{1 + (n_0/n)^2}} \int \sin \theta d\theta d\phi (-K_n \partial_\phi K_n^*) \delta_{q,z}. \end{aligned} \quad (2.55)$$

From Eq. (2.32), the linear momentum flux is given by

$$\begin{aligned}\mathcal{F}_q &= \frac{2\bar{\rho}M^2}{c_s^2} \sum_{n \geq 1} \int \sin \theta \, d\theta \, d\phi \, |K_n|^2 n^q \\ &= \frac{2\bar{\rho}M^2}{c_s^2} \sum_{n \geq 1} \int \sin \theta \, d\theta \, d\phi \, |K_n|^2 \sin \theta \sin \phi \delta_{q,y}.\end{aligned}\tag{2.56}$$

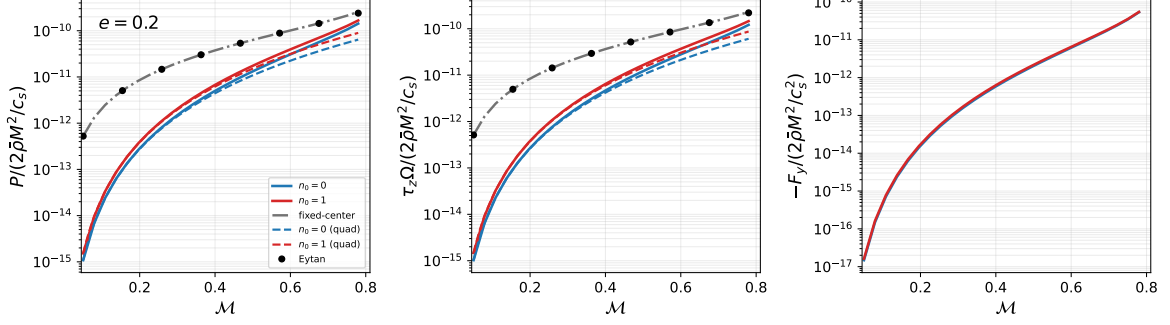


Figure 2. Comparison of the full radiation fluxes with the quadrupole approximation (2.66) in the extreme-mass-ratio limit for $e = 0.2$. The panels show P , $\tau_z \tilde{\Omega}$, and $-F_y$, normalized respectively by $2\bar{\rho}M^2/c_s$, $2\bar{\rho}M^2/c_s$, and $2\bar{\rho}M^2/c_s^2$. The results in the fixed-center non-self-gravitating ($n_0 = 0$) setup are shown by the gray dot-dashed lines, which agree closely with the multipole-orbital-harmonic sums of Ref. [5] (marked as black points).

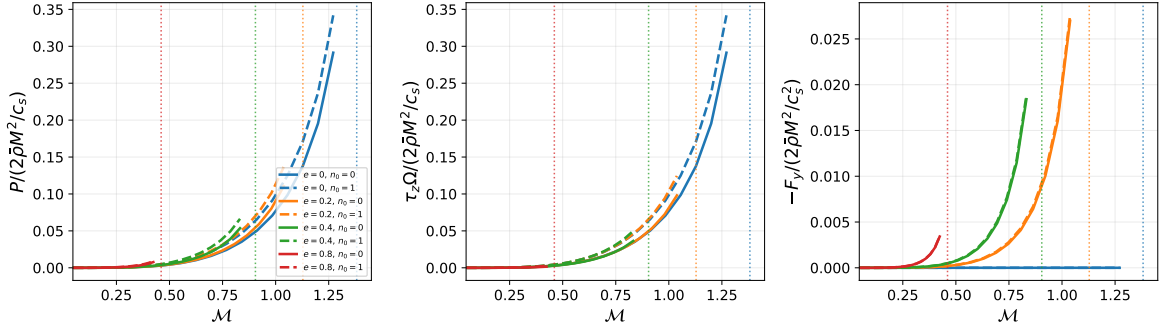


Figure 3. Sound-wave fluxes from eccentric binaries with $\nu = 0.2$ in a homogeneous classical Newtonian fluid. The panels show P , $\tau_z \tilde{\Omega}$, and $-F_y$, normalized respectively by $2\bar{\rho}M^2/c_s$, $2\bar{\rho}M^2/c_s$, and $2\bar{\rho}M^2/c_s^2$. These normalized fluxes depend only on $\{e, \mathcal{M}, n_0, \nu\}$. Solid and dashed curves correspond to $n_0 = 0$ and $n_0 = 1$, respectively. The vertical dotted lines indicate the supersonic threshold (2.59); the rapid growth near these lines signals the onset of UV divergence.

The time-averaged angular-momentum flux is along the z direction, parallel to the orbital angular momentum, whereas the time-averaged linear-momentum flux lies along the instantaneous velocity of the lighter component near periastron, where the emission is strongest. Since the periastron of body 1 is at the positive x -axis and $m_1 \geq m_2$, the linear momentum flux is along the negative y -axis. For equal masses or a circular orbit, $\mathcal{F}_y = 0$. See Figs. 2 and 3 for representative examples in the extreme-mass-ratio limit and for a finite-mass-ratio binary with $\nu = 0.2$, respectively. In Fig. 2, we also compare the full results with the

quadrupole approximation, which is accurate in the deep-subsonic regime. The black points show the fixed-center, non-self-gravitating single-perturber results obtained by evaluating the multipole-harmonic sums of Ref. [5], which were derived using a direct time-domain approach and agree with the corresponding limit of our flux formulae.

2.5 Ultraviolet divergence from stationary phase

Now we examine the UV behavior of the radiation fluxes. Consider one term in the source integral, written in physical time as

$$\mathcal{A}_{nI}(\mathbf{n}) = \frac{1}{T} \int_0^T dt \exp \left[in\tilde{\Omega}t - ik_n \mathbf{n} \cdot \mathbf{X}_I(t) \right]. \quad (2.57)$$

In the large- n limit, $k_n/(n\tilde{\Omega}) \rightarrow 1/c_s$. A real stationary point exists if and only if

$$\frac{d}{dt} \left[n\tilde{\Omega}t - k_n \mathbf{n} \cdot \mathbf{X}_I(t) \right] = 0 \iff \mathbf{n} \cdot \mathbf{v}_I(t) = c_s + O(n^{-2}). \quad (2.58)$$

Hence the source has a high-frequency stationary-phase tail precisely when at least one component becomes locally supersonic somewhere on the orbit, $\max_{I,t} |\mathbf{v}_I(t)| > c_s$. This corresponds to Eq. (2.47) in the case of a Newtonian two-body system in the center-of-mass frame. For a Keplerian binary, $|\mathbf{v}_1 - \mathbf{v}_2|_{\max} = a\tilde{\Omega}\sqrt{\frac{1+e}{1-e}}$. Therefore, the exact point-source criterion is

$$\frac{1 + \sqrt{1 - 4\nu}}{2} \mathcal{M} \sqrt{\frac{1+e}{1-e}} > 1, \quad (2.59)$$

with $\mathcal{M} = a\Omega$. When Eq. (2.59) is not satisfied, the phase has no real stationary point and the Fourier harmonics at large- n are rapidly suppressed. When it is satisfied, stationary phase gives $K_n \sim n^{-1/2}$ over an open set of emission directions. Consequently, $\int d\Omega_{\mathbf{n}} |K_n|^2 \sim n^{-1}$, and the radiation fluxes of point particles acquire a logarithmic dependence on the UV cutoff, indicating the failure of the theory at sufficiently short distances.

2.6 Quadrupole approximation as subsonic limit

If $ak_n \ll 1$, we can use a long-wavelength expansion of $e^{-i\mathbf{k}^{(n)} \cdot \mathbf{X}_I}$ in Eq. (2.26), and the leading term corresponds to the quadrupole radiation. For radiation with a relativistic wave speed, the long-wavelength approximation is justified for nonrelativistic binaries because ak_n is of order the orbital velocity. In the present case, however, $ak_n \sim a/(c_s T) \sim \sqrt{M/a}/c_s \equiv \mathcal{M}$. Thus the effective long-wavelength expansion is justified only in the deep subsonic regime, when the Mach number $\mathcal{M} \ll 1$. In this regime, we can use the quadrupole approximation:

$$\begin{aligned} \frac{n_{(n)}}{4\pi/c_s^2} &= \sum_{I=1,2} \frac{m_I}{T} \int_0^T d\mathbf{t} e^{i\Omega_n \mathbf{t}} e^{-i\mathbf{k}^{(n)} \cdot \mathbf{X}_I} \\ &\approx \frac{n_{(n)}^{[2]}}{4\pi/c_s^2} \equiv - \sum_I \frac{m_I k_n^2}{2} \frac{1}{T} \int_0^T d\mathbf{t} e^{i\Omega_n \mathbf{t}} n^i n^j X_I^i X_I^j \\ &= - \frac{\nu M k_n^2}{2} \mathcal{I}_{ij} n^i n^j, \end{aligned} \quad (2.60)$$

with

$$\mathcal{I}_{ij} = \frac{1}{T} \int_0^T d\mathbf{t} e^{i\Omega_n \mathbf{t}} X^i X^j = \int_0^{2\pi} \frac{d\xi}{2\pi} (1 - e \cos \xi) e^{in(\xi - e \sin \xi)} X^i X^j. \quad (2.61)$$

Note that for $n \geq 1$,

$$|\mathcal{I}_{kk}|^2 + 2\mathcal{I}_{ij}\mathcal{I}_{ij}^* = \frac{4a^4 \left\{ 4e^2 (e^2 - 1) \left[(e^2 - 1) n^2 - 1 \right] (J'_n)^2 - 4e (3e^4 - 7e^2 + 4) n J_n J'_n + \left[3e^4 - 4 (e^2 - 1)^3 n^2 - 4e^2 + 4 \right] (J_n)^2 \right\}}{e^4 n^4}, \quad (2.62)$$

$$\epsilon_{qkl}\mathcal{I}_{ki}^*\mathcal{I}_{li} = \delta_{q,z} \frac{8ia^4\sqrt{1-e^2} \left[(e^2 - 1) n J_n + e J'_n \right] \left[2e (e^2 - 1) n J'_n - (e^2 - 2) J_n \right]}{e^4 n^4}. \quad (2.63)$$

It is useful to separate the reduced quadrupole pattern from the overall source normalization. Define

$$Q_n(\mathbf{n}) \equiv \mathcal{I}_{ij} n^i n^j, \quad n_{(n)}^{[2]} = -\frac{4\pi}{c_s^2} \frac{\nu M k_n^2}{2} Q_n(\mathbf{n}). \quad (2.64)$$

The required angular identities are then

$$\begin{aligned} \int d\Omega_{\mathbf{n}} |Q_n|^2 &= \mathcal{I}_{ij}\mathcal{I}_{kl}^* \int d\Omega_{\mathbf{n}} n^i n^j n^k n^l = \frac{4\pi}{15} (|\mathcal{I}_{kk}|^2 + 2\mathcal{I}_{ij}\mathcal{I}_{ij}^*), \\ \int d\Omega_{\mathbf{n}} \epsilon_{qkl} x^l Q_n \partial_k Q_n^* &= R \mathcal{I}_{ij}\mathcal{I}_{sp}^* \int d\Omega_{\mathbf{n}} \epsilon_{qkl} n^l n^i n^j \partial_k (n^s n^p) = \frac{16\pi}{15} \epsilon_{qkl} \mathcal{I}_{ki}^* \mathcal{I}_{li}. \end{aligned} \quad (2.65)$$

Restoring the prefactor in $n_{(n)}^{[2]}$ in Eqs. (2.29) and (2.34) yields $\mathcal{F}_q = 0$, and

$$\begin{aligned} P &= \sum_{n \geq 1} \frac{2\pi}{15} \frac{\nu^2 M^2 \bar{\rho} k_n^3 \Omega_n}{c_s} (|\mathcal{I}_{kk}|^2 + 2\mathcal{I}_{ij}\mathcal{I}_{ij}^*), \\ \boldsymbol{\tau} &= \tau \mathbf{e}_z = \sum_{n \geq 1} \frac{8\pi}{15} \frac{\nu^2 M^2 \bar{\rho} k_n^3}{c_s^2} (-i \epsilon_{zkl} \mathcal{I}_{ki}^* \mathcal{I}_{li}) \mathbf{e}_z. \end{aligned} \quad (2.66)$$

For a circular orbit ($e = 0$),

$$|\mathcal{I}_{kk}|^2 + 2\mathcal{I}_{ij}\mathcal{I}_{ij}^* = \frac{a^4}{2} (\delta_{n,2} + \delta_{n,-2}), \quad \epsilon_{zkl} \mathcal{I}_{ki}^* \mathcal{I}_{li} = i \frac{a^4}{4} (\delta_{n,2} - \delta_{n,-2}), \quad (2.67)$$

hence

$$P = \frac{2\pi}{15} \frac{\nu^2 M^2 \bar{\rho} k_2^3 \Omega a^4}{c_s} = \tau \tilde{\Omega}. \quad (2.68)$$

This agrees with the long-wavelength limit of Eq. (2.40).

If the fluid self-gravity is neglected by setting $m = 0$, we obtain

$$\begin{aligned} P &= \frac{4\pi}{15} \frac{\nu^2 M^2 \bar{\rho} \Omega^4 a^4}{c_s} \frac{7\sqrt{1-e^2} - 3(1-e^2)}{1-e^2}, \\ \tau &= \frac{16\pi}{15} \frac{\nu^2 M^2 \bar{\rho} \Omega^3 a^4}{c_s^2} \sqrt{1-e^2}. \end{aligned} \quad (2.69)$$

If the binary evolves adiabatically under the associated radiation reaction, the secular rates of change of the orbital elements $\{a, e\}$ are given by

$$\dot{a} = -\frac{2a^2 P}{\nu M^2}, \quad \dot{e} = -\left(\frac{1-e^2}{e} \right) \frac{a}{\nu M^2} \left(P - \frac{\tau \tilde{\Omega}}{\sqrt{1-e^2}} \right). \quad (2.70)$$

For Eq. (2.69) we find that

$$P - \frac{\tau \tilde{\Omega}}{\sqrt{1-e^2}} = \frac{28\pi}{15} \frac{\nu^2 M^2 \bar{\rho} \Omega^4 a^4}{c_s} \frac{e^2 + \sqrt{1-e^2} - 1}{1-e^2} > 0. \quad (2.71)$$

Thus, the sound radiation in this case tends to circularize the orbit.

3 Binary in a quantum fluid

The nonrelativistic limit of a massive fundamental field also admits an effective description as a self-gravitating Newtonian fluid. Consider a real, canonically normalized scalar field ϕ with the potential:

$$V(\phi) = \frac{1}{2}m_\phi^2\phi^2 + \frac{1}{4}\lambda\phi^4, \quad (3.1)$$

where m_ϕ is the scalar mass and λ is the quartic self-coupling. In the nonrelativistic limit, the slow mode of the scalar field in the FLRW background of Eq. (2.1) satisfies the Schrödinger–Poisson (SP) equation [21]:⁶

$$\begin{aligned} \nabla^2\Phi &= 4\pi a^2 (m_\phi|\psi|^2 - \bar{\rho}), \\ i\left(\dot{\psi} + \frac{3}{2}H\psi\right) &= -\frac{1}{2m_\phi a^2}\nabla^2\psi + m_\phi\Phi\psi + \frac{3\lambda}{4m_\phi^2}|\psi|^2\psi, \end{aligned} \quad (3.3)$$

with

$$\phi = \frac{1}{\sqrt{2m_\phi}}(\psi e^{-im_\phi t} + \text{c.c.}). \quad (3.4)$$

The Madelung transformation recasts the wavefunction as a quantum fluid. We introduce the variables $\{\rho, \mathbf{v}\}$ by $\psi \equiv \sqrt{\frac{\rho}{m_\phi}} e^{im_\phi s}$ and $\mathbf{v} \equiv \frac{\nabla s}{a} = \frac{i}{2a\rho}(\psi\nabla\psi^* - \psi^*\nabla\psi)$. The SP equations become

$$\dot{\rho} + 3H\rho + \frac{\nabla \cdot (\rho\mathbf{v})}{a} = 0, \quad (3.5)$$

$$\dot{\mathbf{v}} + H\mathbf{v} + \frac{\mathbf{v} \cdot \nabla \mathbf{v}}{a} = \frac{1}{2m_\phi^2 a^3} \nabla \left(\frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} \right) - \frac{\nabla \left(\Phi + \frac{\rho}{\rho_{\text{int}}} \right)}{a}, \quad (3.6)$$

$$\nabla^2\Phi = 4\pi a^2(\rho - \bar{\rho}), \quad (3.7)$$

where $\rho_{\text{int}} \equiv 4m_\phi^4/(3\lambda)$. A homogeneous background solution is given by $\nabla\bar{\rho} = \nabla\bar{\Phi} = \bar{\mathbf{v}} = \mathbf{0}$. As in Sec. 2, we are interested in the linear perturbation of this fluid background by a set of nonrelativistic point masses, corresponding to the density perturbation ρ_* . We decompose the fields as

$$\rho = \bar{\rho}(1 + \alpha) + \rho_*, \quad \Phi = \bar{\Phi} + \Phi, \quad \mathbf{v} = \bar{\mathbf{v}} + \mathbf{u}_*, \quad (3.8)$$

The linearized fluid equations then read

$$\nabla^2\Phi = 4\pi a^2(\bar{\rho}\alpha + \rho_*), \quad (3.9)$$

$$\dot{\mathbf{u}} + H\mathbf{u} = -\frac{\nabla\Phi}{a} - \frac{\bar{\rho}}{\rho_{\text{int}}a}\nabla\alpha + \frac{1}{4m_\phi^2 a^3}\nabla^3\alpha, \quad (3.10)$$

$$\dot{\alpha} + \frac{\nabla \cdot \mathbf{u}}{a} = 0. \quad (3.11)$$

⁶If we use $\Psi = a^{3/2}\psi$, then

$$\nabla^2\Phi = 4\pi \left(\frac{m_\phi|\Psi|^2}{a} - a^2\bar{\rho} \right), \quad i\dot{\Psi} = -\frac{1}{2m_\phi a^2}\nabla^2\Psi + m_\phi\Phi\Psi + \frac{3\lambda}{4m_\phi^2 a^3}|\Psi|^2\Psi. \quad (3.2)$$

It follows that

$$\ddot{\alpha} + 2H\dot{\alpha} - \left(\frac{\bar{\rho}}{\rho_{\text{int}}}\right) \frac{\nabla^2 \alpha}{a^2} + \frac{1}{4m_\phi^2} \frac{\nabla^4 \alpha}{a^4} = 4\pi(\bar{\rho}\alpha + \rho_*). \quad (3.12)$$

Henceforth we consider the sub-horizon limit with $a = 1$, so the fluid equations reduce to

$$\nabla^2 \Phi = 4\pi(\bar{\rho}\alpha + \rho_*), \quad (3.13)$$

$$\dot{\mathbf{u}} = -\nabla \Phi - \frac{\bar{\rho}}{\rho_{\text{int}}} \nabla \alpha + \frac{1}{4m_\phi^2} \nabla^3 \alpha, \quad (3.14)$$

$$\dot{\alpha} + \nabla \cdot \mathbf{u} = 0, \quad (3.15)$$

and⁷

$$\ddot{\alpha} - \left(\frac{\bar{\rho}}{\rho_{\text{int}}}\right) \nabla^2 \alpha + \frac{1}{4m_\phi^2} \nabla^4 \alpha = 4\pi(\bar{\rho}\alpha + \rho_*). \quad (3.17)$$

3.1 Sound waves

As in the classical-fluid case, we now turn to a Lagrangian formulation. The Lagrangian for $\{s, \alpha, \Phi\}$ can be taken as

$$\mathcal{L} = -\frac{\bar{\rho}}{2} |\nabla s|^2 - \frac{\bar{\rho}^2}{2\rho_{\text{int}}} \alpha^2 - \bar{\rho} \Phi \alpha - \frac{\bar{\rho}}{8m_\phi^2} |\nabla \alpha|^2 + \bar{\rho} s \dot{\alpha} - \frac{1}{8\pi} |\nabla \Phi|^2 - \rho_* \Phi. \quad (3.18)$$

From Noether's theorem (setting $\rho_* = 0$), we obtain the energy flux density:

$$\mathbf{S} = \frac{\partial \mathcal{L}}{\partial \nabla s} \dot{s} + \frac{\partial \mathcal{L}}{\partial \nabla \alpha} \dot{\alpha} + \frac{\partial \mathcal{L}}{\partial \nabla \Phi} \dot{\Phi} = -\bar{\rho} \dot{s} \nabla s - \frac{\bar{\rho}}{4m_\phi^2} \dot{\alpha} \nabla \alpha - \frac{1}{4\pi} \dot{\Phi} \nabla \Phi, \quad (3.19)$$

the energy density:

$$\mathcal{E} = \frac{\partial \mathcal{L}}{\partial \dot{\alpha}} \dot{\alpha} - \mathcal{L} = \bar{\rho} s \dot{\alpha} - \mathcal{L}, \quad (3.20)$$

the linear momentum flux density:

$$\begin{aligned} \Pi_q^i &= - \left[\frac{\partial \mathcal{L}}{\partial (\partial_i s)} \partial_q s + \frac{\partial \mathcal{L}}{\partial (\partial_i \Phi)} \partial_q \Phi + \frac{\partial \mathcal{L}}{\partial (\partial_i \alpha)} \partial_q \alpha - \delta_{iq} \mathcal{L} \right] \\ &= \bar{\rho} \partial_i s \partial_q s + \frac{1}{4\pi} \partial_i \Phi \partial_q \Phi + \frac{\bar{\rho}}{4m_\phi^2} \partial_i \alpha \partial_q \alpha + \delta_{iq} \mathcal{L}, \end{aligned} \quad (3.21)$$

the linear momentum density:

$$p_q = -\frac{\partial \mathcal{L}}{\partial \dot{\alpha}} \partial_q \alpha = -\bar{\rho} s \partial_q \alpha, \quad (3.22)$$

⁷For a scalar field with a nonminimal coupling ($\mathcal{L} \supset -\frac{1}{2}m_\phi^2 \phi^2 - \frac{\xi}{2} R \phi^2$) to gravity [22], the linearized equation takes a similar form, with

$$\ddot{\alpha} - \left(\frac{\bar{\rho}}{\rho_{\text{int}}} + \frac{8\pi\xi\bar{\rho}}{m_\phi^2}\right) \nabla^2 \alpha + \left(1 - \frac{16\pi\xi^2\bar{\rho}}{m_\phi^2}\right) \frac{1}{4m_\phi^2} \nabla^4 \alpha = 4\pi(\bar{\rho}\alpha + \rho_*). \quad (3.16)$$

the angular momentum flux density:

$$\begin{aligned}
j_q^i &= \frac{1}{2} \epsilon_{qkl} \left[\frac{\partial \mathcal{L}}{\partial (\partial_i s)} A_{kl}^j \partial_j s + \frac{\partial \mathcal{L}}{\partial (\partial_i \alpha)} A_{kl}^j \partial_j \alpha + \frac{\partial \mathcal{L}}{\partial (\partial_i \Phi)} A_{kl}^j \partial_j \Phi - A_{kl}^i \mathcal{L} \right] \\
&= \frac{1}{2} \epsilon_{qkl} \left[-\bar{\rho} \partial_i s \left(x^l \partial_k - x^k \partial_l \right) s - \frac{\bar{\rho}}{4m_\phi^2} \partial_i \alpha \left(x^l \partial_k - x^k \partial_l \right) \alpha \right. \\
&\quad \left. - \frac{1}{4\pi} \partial_i \Phi \left(x^l \partial_k - x^k \partial_l \right) \Phi - (\delta_{ik} x^l - \delta_{il} x^k) \mathcal{L} \right],
\end{aligned} \tag{3.23}$$

and the angular momentum density:

$$j_q = \frac{1}{2} \epsilon_{qkl} \left[\frac{\partial \mathcal{L}}{\partial \dot{\alpha}} A_{kl}^j \partial_j \alpha \right] = \bar{\rho} \epsilon_{qkl} x^l s \partial_k \alpha. \tag{3.24}$$

Equation (3.19) also follows directly from the underlying wavefunction theory. Denoting the point-particle mass density by $\rho_*(\mathbf{x}, t) = \sum_I m_I \delta^3(\mathbf{x} - \mathbf{X}_I(t))$, a convenient Newtonian Lagrangian density is

$$\mathcal{L}_\psi = \frac{i}{2} (\psi^* \dot{\psi} - \dot{\psi}^* \psi) - \frac{1}{2m_\phi} |\nabla \psi|^2 - \frac{3\lambda}{8m_\phi^2} |\psi|^4 - m_\phi \Phi |\psi|^2 - \rho_* \Phi - \frac{1}{8\pi} |\nabla \Phi|^2 + \sum_I \frac{m_I}{2} |\dot{\mathbf{X}}_I|^2 \delta^3(\mathbf{x} - \mathbf{X}_I). \tag{3.25}$$

With $\psi = \sqrt{\rho/m_\phi} e^{im_\phi s}$, the spatial-gradient terms give the energy flux

$$\begin{aligned}
\mathbf{S} &= \frac{\partial \mathcal{L}_\psi}{\partial \nabla \psi} \dot{\psi} + \frac{\partial \mathcal{L}_\psi}{\partial \nabla \psi^*} \dot{\psi}^* + \frac{\partial \mathcal{L}_\psi}{\partial \nabla \Phi} \dot{\Phi} \\
&= -\frac{1}{2m_\phi} (\dot{\psi} \nabla \psi^* + \dot{\psi}^* \nabla \psi) - \frac{1}{4\pi} \dot{\Phi} \nabla \Phi \\
&= -\rho \dot{s} \nabla s - \frac{1}{4m_\phi^2} \frac{\dot{\rho} \nabla \rho}{\rho} - \frac{1}{4\pi} \dot{\Phi} \nabla \Phi.
\end{aligned} \tag{3.26}$$

Linearizing $\rho = \bar{\rho}(1 + \alpha)$ reproduces Eq. (3.19).

3.2 Radiation from a localized periodic source

Introducing a new time coordinate $\mathbf{t} = t/(2m_\phi)$, Eq. (3.17) becomes

$$\partial_{\mathbf{t}}^2 \alpha - c_S^2 \nabla^2 \alpha + \nabla^4 \alpha - m^2 \alpha = n(\mathbf{t}, \mathbf{x}) \equiv 16\pi m_\phi^2 \rho_*(t = 2m_\phi \mathbf{t}, \mathbf{x}), \tag{3.27}$$

with

$$m \equiv \sqrt{16\pi m_\phi^2 \bar{\rho}}, \quad c_S^2 \equiv \frac{4m_\phi^2 \bar{\rho}}{\rho_{\text{int}}}. \tag{3.28}$$

The dispersion relation of a free plane wave $\alpha \propto e^{i(\mathbf{k} \cdot \mathbf{x} - \omega \mathbf{t})}$ is $\omega^2 = c_S^2 |\mathbf{k}|^2 + |\mathbf{k}|^4 - m^2$. For a repulsive quartic self-interaction ($\rho_{\text{int}} = 4m_\phi^4/(3\lambda) > 0$) and negligible quantum pressure (i.e., neglecting the ∇^4 term in the wave equation⁸), the problem is similar to that of a classical fluid considered in Sec. 2; see Eq. (2.21). For an attractive or vanishing quartic self-interaction, the quantum pressure is always important for radiation.

⁸This approximation is justified for radiation modes satisfying $k \ll c_S$ (corresponding to wavelengths much longer than the healing length $\xi_h = \sqrt{2}/c_S$), so that the quantum-pressure term is negligible.

In the frequency domain, Eq. (3.27) reads $(\nabla^4 - c_S^2 \nabla^2 - m^2 - \omega^2) \tilde{\alpha} = \tilde{n}$. For positive frequency $\omega > 0$, the outgoing solution to $(\nabla^4 - c_S^2 \nabla^2 - m^2 - \omega^2) f(\mathbf{x}, \mathbf{x}') = \delta^3(\mathbf{x} - \mathbf{x}')$ is

$$\begin{aligned} f(\mathbf{x}, \mathbf{x}') &= \int \frac{d^3 k}{(2\pi)^3} \frac{e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')}}{|\mathbf{k}|^4 + c_S^2 |\mathbf{k}|^2 - m^2 - (\omega + i0^+)^2} \\ &= \frac{e^{iq_+ |\mathbf{x} - \mathbf{x}'|} - e^{-q_- |\mathbf{x} - \mathbf{x}'|}}{4\pi(q_+^2 + q_-^2) |\mathbf{x} - \mathbf{x}'|} \\ &\xrightarrow{|\mathbf{x} - \mathbf{x}'| \rightarrow \infty} \frac{1}{\sqrt{c_S^4 + 4(\omega^2 + m^2)}} \frac{e^{iq_+ |\mathbf{x} - \mathbf{x}'|}}{4\pi |\mathbf{x} - \mathbf{x}'|}, \end{aligned} \quad (3.29)$$

where

$$q_{\pm} = \sqrt{\frac{\mp c_S^2 + \sqrt{c_S^4 + 4(\omega^2 + m^2)}}{2}} > 0. \quad (3.30)$$

In the limit $(\omega^2 + m^2)/c_S^4 \rightarrow 0$, $q_+ \rightarrow \sqrt{\omega^2 + m^2}/c_S$, $f(\mathbf{x}, \mathbf{x}') \rightarrow \frac{1}{c_S^2} \frac{e^{i(\sqrt{\omega^2 + m^2}/c_S) |\mathbf{x} - \mathbf{x}'|}}{4\pi |\mathbf{x} - \mathbf{x}'|}$, which is as expected because the wave equation in this limit becomes $[\nabla^2 + (m^2 + \omega^2)/c_S^2] \tilde{\alpha} \approx -\tilde{n}/c_S^2$ (see Eq. (2.24)).

For the remainder of this section, we set $c_S = 0$; retaining a nonzero c_S only changes the dispersion relation and the corresponding mode normalization. The radiation field generated by a source with physical period $T \equiv 2\pi/\tilde{\Omega}$, or auxiliary-time period $\mathbf{T} \equiv T/(2m_\phi) = 2\pi/\Omega$, is given by

$$\alpha(\mathbf{t}, \mathbf{x} = R\mathbf{n}) = \frac{1}{8\pi R} \sum_n \frac{n_{(n)}}{k_n^2} e^{i[\mathbf{k}^{(n)} \cdot \mathbf{x} - \Omega_n \mathbf{t}]}, \quad (3.31)$$

$$n_{(n)} = \int d^3 x e^{-i\mathbf{k}^{(n)} \cdot \mathbf{x}} \left[\frac{1}{\mathbf{T}} \int_0^{\mathbf{T}} d\mathbf{t} n(\mathbf{t}, \mathbf{x}) e^{i\Omega_n \mathbf{t}} \right], \quad (3.32)$$

where $\Omega_n \equiv n\Omega$, $\Omega = 2m_\phi \tilde{\Omega}$, and the two-sided expansion uses the signed convention $\mathbf{k}^{(n)} = k_n \mathbf{n}$, with $k_n = \text{sgn}(n)(\Omega_n^2 + m^2)^{1/4}$ for $n \neq 0$. Note that $n_0 \equiv m/\Omega = \sqrt{4\pi\bar{\rho}}/\tilde{\Omega}$. We also have

$$\begin{pmatrix} \Phi \\ s \end{pmatrix} = \frac{1}{8\pi R} \sum_n \begin{pmatrix} \kappa_n^\Phi \\ \kappa_n^s \end{pmatrix} \frac{n_{(n)}}{k_n^2} e^{i[\mathbf{k}^{(n)} \cdot \mathbf{x} - \Omega_n \mathbf{t}]}, \quad (3.33)$$

with

$$\kappa_n^\Phi = \frac{-4\pi\bar{\rho}}{k_n^2}, \quad \kappa_n^s = \frac{-i\Omega_n/(2m_\phi)}{k_n^2}. \quad (3.34)$$

Using the far-zone fields above, the time-averaged energy flux is

$$P = R^2 \int d\Omega_{\mathbf{n}} \mathbf{S} \cdot \mathbf{n} = \int d\Omega_{\mathbf{n}} \sum_{n \geq 1} \frac{\bar{\rho}\Omega_n}{128\pi^2 m_\phi^3 k_n^3} |n_{(n)}|^2. \quad (3.35)$$

The time-averaged energy density is given by

$$\mathcal{E} = \frac{\bar{\rho}}{2} |\nabla s|^2 + \bar{\rho} \Phi \alpha + \frac{\bar{\rho}}{8m_\phi^2} |\nabla \alpha|^2 + \frac{1}{8\pi} |\nabla \Phi|^2 = \frac{1}{R^2} \sum_{n \geq 1} \frac{\bar{\rho}\Omega_n^2}{128\pi^2 m_\phi^2 k_n^6} |n_{(n)}|^2. \quad (3.36)$$

This quantity satisfies $\mathbf{S} \cdot \mathbf{n} = \mathcal{E}v_g$ mode by mode, where $v_g = k_n^3/(m_\phi\Omega_n)$ is the group velocity of the sound wave. The time-averaged linear momentum density is

$$p_q = -\bar{\rho}s\partial_q\alpha = \frac{1}{R^2} \sum_{n \geq 1} \frac{\bar{\rho}\Omega_n}{64\pi^2 m_\phi k_n^5} |n_{(n)}|^2 n^q, \quad (3.37)$$

so the time-averaged linear momentum flux is

$$\mathcal{F}_q = \sum_{n \geq 1} \int d\Omega_{\mathbf{n}} \frac{\bar{\rho}}{64\pi^2 m_\phi^2 k_n^2} |n_{(n)}|^2 n^q, \quad (3.38)$$

which is consistent with the direct evaluation: $\mathcal{F}_q = R^2 \int d\Omega_{\mathbf{n}} \Pi_q^i n^i$. The time-averaged angular momentum density is

$$j_q = \bar{\rho}\epsilon_{qkl}x^l s\partial_k\alpha = \frac{1}{R^2} \sum_n (-i) \frac{\bar{\rho}\Omega_n}{128\pi^2 m_\phi k_n^6} \epsilon_{qkl}x^l n_{(n)} \partial_k n_{(n)}^*. \quad (3.39)$$

The time-averaged angular momentum flux is

$$\tau_q = \text{Re} \int d\Omega_{\mathbf{n}} \sum_{n \geq 1} (-i) \frac{\bar{\rho}}{64\pi^2 m_\phi^2 k_n^3} \epsilon_{qkl}x^l n_{(n)} \partial_k n_{(n)}^*, \quad (3.40)$$

which is consistent with the direct evaluation: $\tau_q = R^2 \int d\Omega_{\mathbf{n}} j_q^i n^i$.

3.3 A common structure of fluxes

It is useful to isolate the kinematical structure common to the radiation fluxes from a localized periodic source. Using a positive-frequency convention, define $\tilde{\Omega}_n \equiv n\tilde{\Omega}$ and choose

$$A_n^{\text{cl}} = \frac{\bar{\rho}c_s^3}{8\pi^2} \frac{\Omega_n}{k_n}, \quad A_n^{\text{q}} = \frac{\bar{\rho}\Omega_n}{128\pi^2 m_\phi^3 k_n^3}, \quad (3.41)$$

for the classical and quantum fluids (with $c_s = 0$), respectively. The time-averaged energy, linear-momentum, and angular-momentum fluxes then take the common form

$$\begin{pmatrix} P \\ \mathcal{F}_q \\ \tau_q \end{pmatrix} = \sum_{n \geq 1} \int d\Omega_{\mathbf{n}} A_n \begin{pmatrix} |n_{(n)}|^2 \\ \frac{k_n}{\tilde{\Omega}_n} |n_{(n)}|^2 n^q \\ \text{Re} \left[\frac{-i}{\tilde{\Omega}_n} \epsilon_{qkl} x^l n_{(n)} \partial_k n_{(n)}^* \right] \end{pmatrix}. \quad (3.42)$$

This also holds for the Klein–Gordon field reviewed in Appendix B, for which $A_n = \frac{\Omega_n^2}{8\pi^2} \left(1 - \frac{n_0^2}{n^2}\right)^{1/2}$. The common structure indicates that, once the outgoing amplitude and dispersion relation of the scalar radiative mode are known, the conserved charges carried by each mode are fixed kinematically.

3.4 Circular orbit

For the circular binary introduced in Sec. 2.3, Eq. (3.35) gives the energy flux:

$$P = \frac{4\pi\bar{\rho}M^2m_\phi}{\sqrt{\Omega}} \sum_{n \geq 1} \frac{nI_n}{(n^2 + n_0^2)^{3/4}} \equiv \frac{2\bar{\rho}M^2m_\phi}{\sqrt{\Omega}} \mathcal{X}_Q, \quad (3.43)$$

where I_n is given by Eq. (2.41). It is convenient in the quantum-fluid problem to use the dimensionless parameter

$$\mathcal{M}_Q \equiv a\sqrt{\Omega}. \quad (3.44)$$

For an equal-mass circular binary, one has $k_na = \mathcal{M}_Q(n^2 + n_0^2)^{1/4}$, and hence $\mathcal{X}_Q$ depends on $\mathcal{M}_Q$ and n_0 ; in particular, for $m = 0$, $k_1a = \mathcal{M}_Q$. Fig. 4 shows the results for $n_0 = 0$ and $n_0 = 1$.

Unlike the classical-fluid flux, the quantum-fluid flux is UV finite because of the k^4 dispersion introduced by quantum pressure. To see this, we can repeat the analysis in Sec. 2.5. In the large- n limit, $k_n/(n\tilde{\Omega}) \rightarrow 2m_\phi/\sqrt{n\Omega}$. For a point source, a real stationary point exists if and only if

$$\frac{d}{dt} [n\tilde{\Omega}t - k_n\mathbf{n} \cdot \mathbf{X}_I(t)] = 0 \iff \mathbf{n} \cdot \mathbf{v}_I(t) \sim n\tilde{\Omega}/k_n \propto n^{1/2}. \quad (3.45)$$

No finite orbital velocity can satisfy this condition for arbitrarily large n . Hence the large- n tail has no real stationary point and the point-source fluxes are UV convergent.

For an equal-mass circular orbit with $n_0 = 0$, asymptotic analysis shows that

$$\lim_{\mathcal{M}_Q \rightarrow \infty} \mathcal{X}_Q \rightarrow 2\pi \frac{2 \ln \mathcal{M}_Q - 3 \ln 2 + \gamma_E}{\mathcal{M}_Q}, \quad (3.46)$$

where γ_E is the Euler–Mascheroni constant. As $\mathcal{M}_Q$ increases, the oscillations in $\mathcal{X}_Q(\mathcal{M}_Q)$ are damped, while the local mean approaches the asymptotic expression in Eq. (3.46), as shown in Fig. 4.

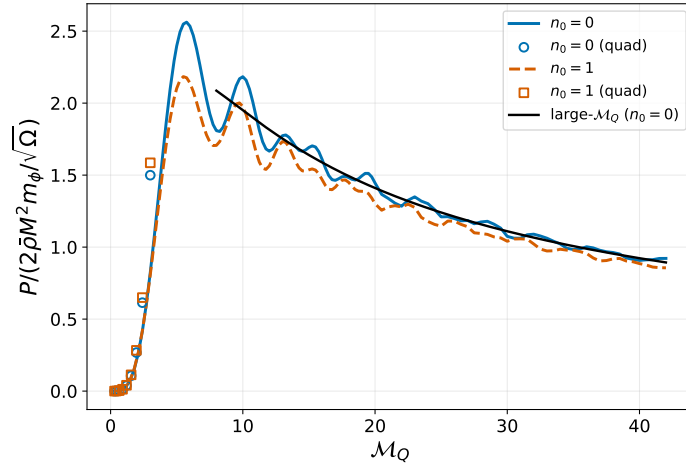


Figure 4. The dimensionless circular-orbit function $\mathcal{X}_Q(\mathcal{M}_Q)$ for an equal-mass binary, shown for $n_0 = 0$ and $n_0 = 1$. The markers show the quadrupole approximation in Eq. (3.51), and the black curve gives the large- $\mathcal{M}_Q$ asymptotic result in Eq. (3.46) for $n_0 = 0$.

3.5 Eccentric orbit

For the eccentric binary introduced in Sec. 2.4, Eqs. (3.35), (3.40), and (3.38) give the radiation fluxes:

$$P = \frac{2\bar{\rho}M^2m_\phi}{\sqrt{\Omega}} \sum_{n \geq 1} \frac{\int \sin \theta d\theta d\phi |K_n|^2 n}{(n^2 + n_0^2)^{3/4}}, \quad (3.47)$$

$$\tau_q \tilde{\Omega} = \frac{2\bar{\rho}M^2m_\phi}{\sqrt{\Omega}} \text{Re} \sum_{n \geq 1} \frac{(-i)}{(n^2 + n_0^2)^{3/4}} \int \sin \theta d\theta d\phi (-K_n \partial_\phi K_n^*) \delta_{q,z}, \quad (3.48)$$

$$\mathcal{F}_q(\sqrt{\Omega}/m_\phi) = \frac{2\bar{\rho}M^2m_\phi}{\sqrt{\Omega}} \sum_{n \geq 1} \int \sin \theta d\theta d\phi \frac{2|K_n|^2}{\sqrt{n^2 + n_0^2}} \sin \theta \sin \phi \delta_{q,y}. \quad (3.49)$$

Examples are shown in Fig. 5 for $\nu = 0.2$; they illustrate the dependence of the fluxes on the eccentricity, the parameter $n_0 = \sqrt{4\pi\bar{\rho}}/\tilde{\Omega}$, and the parameter $\mathcal{M}_Q = a\sqrt{\Omega}$.

As in Sec. 2.6, the quadrupole approximation is valid when $\mathcal{M}_Q = a\sqrt{\Omega} \sim ak_1$ is small. In this approximation, $\mathcal{F}_q = 0$, and

$$P = \sum_{n \geq 1} \frac{2\pi}{15} \nu^2 M^2 \bar{\rho} m_\phi k_n \Omega_n (|\mathcal{I}_{kk}|^2 + 2\mathcal{I}_{ij}\mathcal{I}_{ij}^*), \quad (3.50)$$

$$\boldsymbol{\tau} = \tau \mathbf{e}_z = \sum_{n \geq 1} \frac{16\pi}{15} \nu^2 M^2 \bar{\rho} m_\phi^2 k_n (-i\epsilon_{zkl}\mathcal{I}_{ki}^*\mathcal{I}_{li}) \mathbf{e}_z.$$

For a circular orbit ($e = 0$),

$$P = \frac{2\pi}{15} \nu^2 M^2 \bar{\rho} m_\phi k_2 \Omega a^4 = \tau \tilde{\Omega}. \quad (3.51)$$

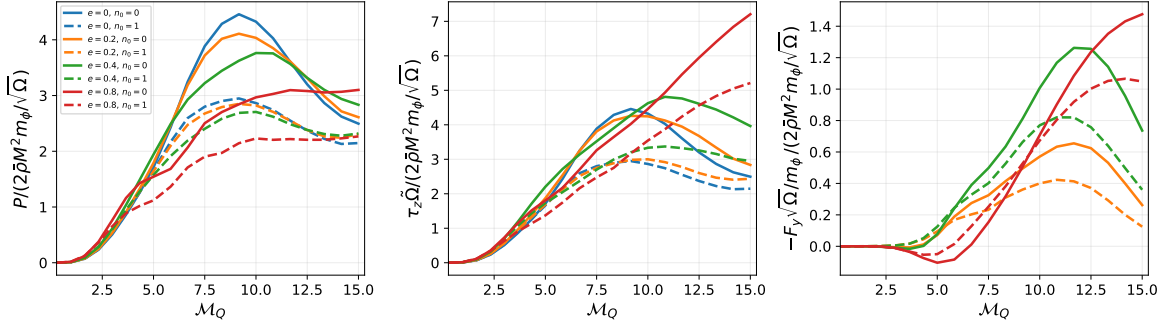


Figure 5. Sound-wave fluxes from eccentric binaries with $\nu = 0.2$ in a homogeneous Newtonian quantum fluid. The panels show P , $\tau_z \tilde{\Omega}$, and $-\mathcal{F}_y \sqrt{\Omega}/m_\phi$, normalized by $2\bar{\rho}M^2m_\phi/\sqrt{\Omega}$. These normalized fluxes depend only on $\{e, \mathcal{M}_Q, n_0, \nu\}$. Solid and dashed curves correspond to $n_0 = 0$ and $n_0 = 1$, respectively.

4 Estimated constraints on the fluid environments of pulsar binaries

We now give an illustrative estimate using binary-pulsar timing, which has long been used to constrain nonstandard orbital perturbations induced by dark environments [23, 24]. The

purpose of this section is not to perform a full timing analysis, but to translate the absence of excess orbital decay into an order-of-magnitude upper bound on the density of a homogeneous fluid background.

At leading Newtonian order, a binary pulsar is described by an eccentric Keplerian orbit with semimajor axis a and orbital period $T = 2\pi/\tilde{\Omega}$. Using

$$E_{\text{orb}} = -\frac{m_1 m_2}{2a}, \quad \tilde{\Omega}^2 = \frac{M}{a^3}, \quad (4.1)$$

the adiabatic energy-balance equation $\dot{E}_{\text{orb}} = -(P_{\text{gw}} + P_{\text{med}})$ gives

$$\dot{T} = -\frac{6\pi a^{5/2}}{m_1 m_2 \sqrt{M}} (P_{\text{gw}} + P_{\text{med}}), \quad (4.2)$$

where P_{med} denotes the additional energy flux carried by waves in the surrounding medium. The leading-order energy flux of gravitational radiation is [25, 26]

$$P_{\text{gw}} = \frac{32}{5} \nu^2 M^2 a^4 \tilde{\Omega}^6 \frac{1 + \frac{73}{24} e^2 + \frac{37}{96} e^4}{(1 - e^2)^{7/2}}. \quad (4.3)$$

We specialize to a homogeneous Newtonian quantum-fluid background with $c_S = 0$ and identify P_{med} with the sound-wave energy flux given by Eq. (3.47), which is both IR finite and UV convergent. For the observational comparison, we write

$$\dot{T}^{\text{th}} = \dot{T}^{\text{GR}} + \dot{T}^{\text{med}}, \quad \dot{T}^{\text{med}} = -\frac{6\pi a^{5/2}}{m_1 m_2 \sqrt{M}} P_{\text{med}} \leq 0, \quad (4.4)$$

where $\dot{T}^{\text{GR}}$ is given by Eq. (4.2) with $P_{\text{med}} = 0$. We define $\Delta\dot{T} \equiv \dot{T}^{\text{meas}} - \dot{T}^{\text{GR}}$. Because an additional dissipative channel makes $\dot{T}$ more negative, we require the theoretical prediction not to fall below the lower 2σ observational limit,

$$\dot{T}^{\text{GR}} + \dot{T}^{\text{med}} \geq \dot{T}^{\text{meas}} - 2\sigma_{\dot{T}^{\text{meas}}}. \quad (4.5)$$

Equivalently,

$$-\dot{T}^{\text{med}} = \frac{6\pi a^{5/2}}{m_1 m_2 \sqrt{M}} P_{\text{med}} \leq B_{2\sigma}, \quad B_{2\sigma} \equiv \max\left(0, 2\sigma_{\dot{T}^{\text{meas}}} - \Delta\dot{T}\right). \quad (4.6)$$

We adopt the $n_0 = 0$ approximation when extracting the bounds. Substituting the resulting upper limits on $\bar{\rho}$ back into $n_0 = \sqrt{4\pi\bar{\rho}}/\tilde{\Omega}$ confirms that the effect of a nonzero n_0 is negligible throughout the parameter range shown. We consider PSR J0737–3039A/B and PSR B1913+16, using the measured binary parameters and orbital-period derivatives listed in Table 1. The resulting upper limits on $\bar{\rho}$ are shown in Fig. 6.

5 Summary and discussion

We have formulated the linear response of a homogeneous Newtonian fluid background to a Keplerian binary as a far-zone radiation problem. In this formulation, the prescribed binary acts as a localized periodic source, and the angular amplitude of the far-zone density perturbation determines the energy, linear-momentum, and angular-momentum fluxes through

Table 1. Binary parameters and orbital-period measurements used to constrain the homogeneous quantum-fluid density. The directly fitted period derivative is denoted by $\dot{T}^{\text{obs}}$, whereas $\dot{T}^{\text{meas}}$ is the corrected value used in the damping test. For PSR J0737–3039A/B, the spin-down mass-loss contribution of pulsar A has been subtracted; for PSR B1913+16, the Galactic correction has been applied. The last two rows give $\Delta\dot{T} = \dot{T}^{\text{meas}} - \dot{T}^{\text{GR}}$ and the one-sided extra-loss budget defined in Eq. (4.6). Derived quantities are evaluated using the unrounded input values before rounding to the displayed precision.

Quantity	PSR J0737–3039A/B	PSR B1913+16
Reference	[27]	[28]
$m_1 [M_\odot]$	1.338185	1.438
$m_2 [M_\odot]$	1.248868	1.390
ν	0.249702	0.249928
T [d]	0.1022515592973	0.322997448918
e	0.087777023	0.6171340
$\dot{T}^{\text{obs}} [10^{-12}]$	−1.247920	−2.423
$\sigma_{\dot{T}^{\text{obs}}} [10^{-15}]$	0.078	1
$\dot{T}^{\text{meas}} [10^{-12}]$	−1.247782	−2.398
$\sigma_{\dot{T}^{\text{meas}}} [10^{-15}]$	0.079	4
$\dot{T}^{\text{GR}} [10^{-12}]$	−1.247827	−2.40263
$\Delta\dot{T} [10^{-15}]$	0.046	4.63
$B_{2\sigma} [10^{-15}]$	0.112	3.37

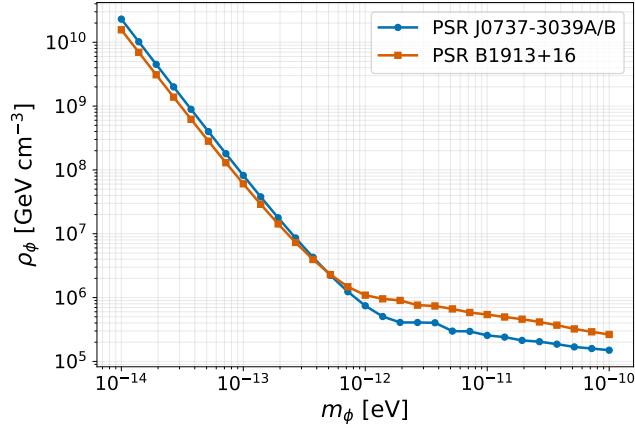


Figure 6. Estimated upper bounds on the homogeneous Newtonian quantum-fluid background density as a function of the scalar mass m_ϕ , obtained from the orbital-decay budgets in Table 1. The two curves correspond to PSR J0737–3039A/B and PSR B1913+16.

Noether currents. This viewpoint is complementary to the usual wake-force description: it gives the radiated sound wave and the global balance laws in a single calculation, while force components that do not enter the dissipative balance must still be extracted from the near-zone wake.

For a classical barotropic fluid, retaining the self-gravity of density perturbations gives a Jeans-modified acoustic mode, with the n th orbital harmonic propagating at $k_n = \sqrt{\Omega_n^2 + m^2}$. We derived flux formulae for localized periodic sources and explicit harmonic representations

for circular and eccentric binaries, with the quadrupole formulae emerging as the controlled deep-subsonic limit. In a nondispersive fluid, the point-source harmonic sum becomes UV-sensitive precisely when a component becomes supersonic, so a physical calculation must introduce a short-distance smoothing scale. By contrast, in a Schrödinger–Poisson fluid, quantum pressure changes the high-frequency scaling to $k_n \propto n^{1/2}$ and makes the point-source fluxes UV convergent. Apart from the dispersion relation and mode normalization, the classical and quantum fluxes share the same kinematical structure. The binary-pulsar estimates above are included only as an illustration of this dependence. More generally, the radiation formulation makes explicit that the dissipative response is not a universal property of the binary alone, but is controlled by the density and dispersion of the background medium. This makes it questionable to use the homogeneous-background result naively as a local formula in an arbitrary inhomogeneous medium.

A Time-domain derivation of the frequency-domain waveform

Here we provide an explicit derivation of the far-zone waveform using the time-domain retarded Green’s function for a real massive Klein-Gordon field in flat spacetime. The retarded Green’s function satisfying $(\square_x - m_\phi^2)G(x, x') = -\delta^4(x - x')$ with $\square = -\partial_t^2 + \nabla^2$ and $m_\phi^2 > 0$ is [29]

$$G(x, x') = \frac{\delta(t' - (t - |\mathbf{x} - \mathbf{x}'|))}{4\pi|\mathbf{x} - \mathbf{x}'|} - \Theta((t - |\mathbf{x} - \mathbf{x}'|) - t') \frac{m_\phi J_1(m_\phi \sqrt{-(x - x')^2})}{4\pi \sqrt{-(x - x')^2}}, \quad (\text{A.1})$$

where the tail term is supported inside the past light cone. The solution of $(\square - m_\phi^2)\phi = -n(t, \mathbf{x})$ is given by

$$\begin{aligned} \phi(t, \mathbf{x}) &= \int d^3x' \frac{n(t - |\mathbf{x} - \mathbf{x}'|, \mathbf{x}')}{4\pi|\mathbf{x} - \mathbf{x}'|} \\ &\quad - \int d^3x' \int_{-\infty}^{t-|\mathbf{x}-\mathbf{x}'|} dt' \frac{m_\phi J_1\left(m_\phi \sqrt{(t-t')^2 - |\mathbf{x} - \mathbf{x}'|^2}\right) n(t', \mathbf{x}')}{4\pi \sqrt{(t-t')^2 - |\mathbf{x} - \mathbf{x}'|^2}}. \end{aligned} \quad (\text{A.2})$$

The source can be written as

$$n(t', \mathbf{x}') = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} n(\omega, \mathbf{x}') e^{-i\omega t'}. \quad (\text{A.3})$$

If $R = |\mathbf{x}|$ is sufficiently large, we have

$$\begin{aligned} & -\frac{m_\phi}{4\pi} \int_{-\infty}^{t-r} dt' \frac{J_1\left(m_\phi \sqrt{(t-t')^2 - r^2}\right)}{\sqrt{(t-t')^2 - r^2}} e^{-i\omega t'} \\ &= -\frac{1}{4\pi r} \left\{ \int_0^\infty dz \frac{J_1(z) e^{i\omega r \sqrt{1+[z/(m_\phi r)]^2}}}{\sqrt{1+[z/(m_\phi r)]^2}} \right\} e^{-i\omega t} \\ &\approx -\frac{1}{4\pi r} \left(e^{i\omega r} - e^{ir \operatorname{sgn}(\omega) \sqrt{\omega^2 - m_\phi^2}} \right) e^{-i\omega t}, \end{aligned} \quad (\text{A.4})$$

where $r = |\mathbf{x} - \mathbf{x}'|$. Here we used the result [29]:

$$\lim_{r \rightarrow \infty} \int_0^\infty dz \frac{J_1(z) e^{i\omega r \sqrt{1+[z/(m_\phi r)]^2}}}{\{1+[z/(m_\phi r)]^2\}^{n/2}} \rightarrow e^{i\omega r} - \left[1 - \left(\frac{m_\phi}{\omega} \right)^2 \right]^{\frac{n-1}{2}} e^{ir \operatorname{sgn}(\omega) \sqrt{\omega^2 - m_\phi^2}}, \quad (\text{A.5})$$

for $\omega^2 > m_\phi^2$. Equation (A.2) can then be written as

$$\begin{aligned}\phi(t, \mathbf{x}) &\approx \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega t} \int d^3x' n(\omega, \mathbf{x}') \left\{ \frac{e^{i\omega r}}{4\pi r} - \frac{1}{4\pi r} \left[e^{i\omega r} - e^{ir \operatorname{sgn}(\omega) \sqrt{\omega^2 - m_\phi^2}} \right] \right\} \\ &= \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \int d^3x' n(\omega, \mathbf{x}') \frac{e^{i(r \operatorname{sgn}(\omega) \sqrt{\omega^2 - m_\phi^2} - \omega t)}}{4\pi r},\end{aligned}\tag{A.6}$$

where the ω integral is restricted to $\omega^2 > m_\phi^2$. For $\mathbf{x} = R\mathbf{n}$, using $r \approx R - \mathbf{n} \cdot \mathbf{x}'$, the field can be written as

$$\phi(t, \mathbf{x}) \approx \frac{1}{4\pi R} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \left[\int d^3x' e^{-i\mathbf{k}_\omega \cdot \mathbf{x}'} n(\omega, \mathbf{x}') \right] e^{i(\mathbf{k}_\omega \cdot \mathbf{x} - \omega t)},\tag{A.7}$$

with $\mathbf{k}_\omega = \operatorname{sgn}(\omega) \sqrt{\omega^2 - m_\phi^2} \mathbf{n}$. For $m_\phi^2 < 0$, replacing $m_\phi \rightarrow im_\phi$ gives $\mathbf{k}_\omega = \operatorname{sgn}(\omega) \sqrt{\omega^2 + m_\phi^2} \mathbf{n}$.

B Radiation of a Klein–Gordon field

We consider a real massive Klein–Gordon field $\phi(t, \mathbf{x})$ coupled to an external source $n(t, \mathbf{x})$ in flat spacetime and described by the Lagrangian:

$$\mathcal{L} = -\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m_\phi^2\phi^2 + n\phi.\tag{B.1}$$

The wave equation reads

$$(\partial_t^2 - \nabla^2 + m_\phi^2)\phi = n.\tag{B.2}$$

For a localized periodic source with $n(t, \mathbf{x}) = n(t + T, \mathbf{x})$, the radiation field is given by

$$\phi(t, \mathbf{x} = R\mathbf{n}) = \frac{1}{4\pi R} \sum_{|n| \geq n_0} n_{(n)} e^{i[\mathbf{k}^{(n)} \cdot \mathbf{x} - \Omega_n t]},\tag{B.3}$$

$$n_{(n)} = \int d^3x e^{-i\mathbf{k}^{(n)} \cdot \mathbf{x}} \left[\frac{1}{T} \int_0^T dt n(t, \mathbf{x}) e^{i\Omega_n t} \right],\tag{B.4}$$

where $n_0 = m_\phi/\Omega$, $\Omega = 2\pi/T$, and $\mathbf{k}^{(n)} = k_n \mathbf{n}$, with $k_n = \operatorname{sgn}(n) \sqrt{\Omega_n^2 - m_\phi^2}$. The corresponding time-averaged radiation fluxes are [20]

$$\begin{aligned}P &= \sum_{n \geq n_0} \frac{\Omega_n^2}{8\pi^2} \left(1 - \frac{n_0^2}{n^2} \right)^{1/2} \int d\Omega_{\mathbf{n}} |n_{(n)}|^2, \\ \mathcal{F}_q &= \sum_{n \geq n_0} \frac{\Omega_n^2}{8\pi^2} \left(1 - \frac{n_0^2}{n^2} \right) \int d\Omega_{\mathbf{n}} |n_{(n)}|^2 n^q, \\ \tau_q &= \operatorname{Re} \sum_{n \geq n_0} (-i) \frac{k_n}{8\pi^2} \int d\Omega_{\mathbf{n}} \epsilon_{qkl} x^l n_{(n)} \partial_k n_{(n)}^*.\end{aligned}\tag{B.5}$$

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